

VLOGS: Virtual Laboratory Observation Tool for Monitoring a Group of Students

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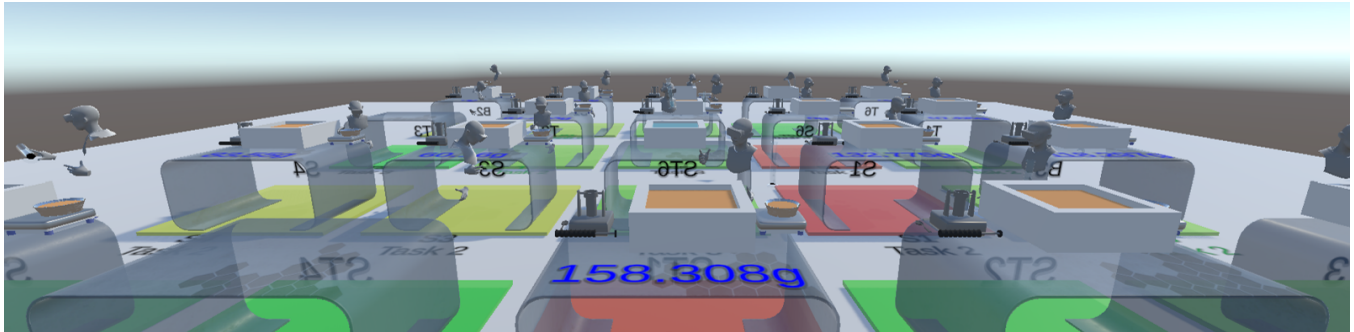


Figure 1: VLOGS: Virtual Laboratory Observation Tool for Monitoring a Group of Students.

ABSTRACT

Virtual laboratories (VLs) enable students to conduct lab experiments in the virtual world using Virtual Reality (VR) technology, providing benefits in areas such as availability, safety as well as scalability. While there are existing platforms that provide VLs with rich content as well as research works on promoting effective learning in VLs, less attention has been paid on VLs from a teaching perspective. Students usually learn and practice in VL sessions with limited help from the instructors. Instructors, on the other hand, could only receive limited information on the performance of the students and could not provide timely feedback to facilitate students' learning. In this work, we present a prototype virtual laboratory monitoring tool, created using a design thinking approach, for addressing teaching needs when conducting a virtual laboratory session simultaneously for multiple students, similar to that in a physical lab environment.

CCS CONCEPTS

• **Applied computing** → **Computer-assisted instruction**; • **Human-centered computing** → **Visualization toolkits**.

KEYWORDS

VR in Education, Teaching and Learning, Laboratory, Virtual Reality

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1 INTRODUCTION

Virtual laboratories (VLs) have been gaining higher utilization in different subject areas of the academic field in recent years [1]. Different methodologies on promoting effective learning in VLs are also actively researched such as that presented in [3]. There are also commercial platforms available for delivering high quality VLs, such as Labster (<https://www.labster.com/>) and VR Lab Academy (<https://www.vrlabacademy.com/>). While effort has been spent on improving the content, learning methodology as well as the infrastructure of VLs, instructors' involvement when conducting a VL session is often limited. Students mostly perform the VLs with limited help. Instructors could only receive limited information on each student's performance and cannot provide just-in-time feedback to guide the students. This motivated us to create a novel VL observation tool, using a design thinking approach [2], for addressing teaching needs when conducting VL sessions with multiple students, similar to that in a physical lab environment.

2 DESIGN PROCESS

A user-centric design thinking framework is adopted to design the VL observation tool on the the unconfined compression test VL in the field of civil engineering [3]. Similar design thinking frameworks have been used to develop instructor tools to facilitate different teaching scenarios such as that in [4]. The framework

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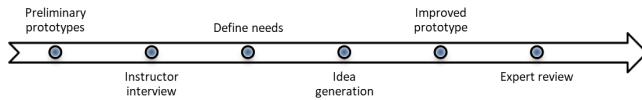


Figure 2: The 6 stages of the adopted design thinking framework to design the VL observation tool.

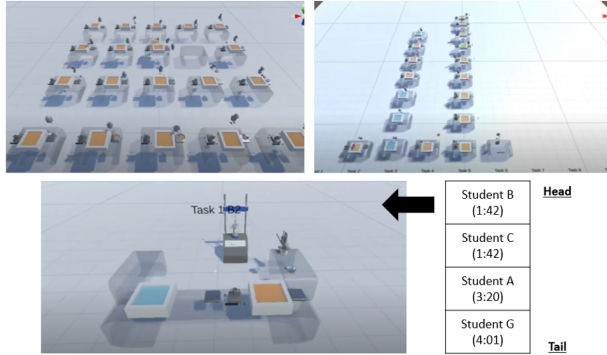


Figure 3: The Grid-based design (top-left) and the Progress-based design (top-right) for monitoring VLs with multiple students. The Queue-based design (bottom) for grading students in VLs.

consists of six stages (Figure 2). At stage 1, three different VL observation preliminary prototypes were created by considering two personas when observing a VL: a monitor and a grader (Figure 3). These preliminary prototypes were then used as instruments to collect feedback from instructors in semi-structured interviews in stage 2. Ten instructors, all from the field of civil engineering and with experience in teaching the unconfined compression test, were invited to participate in the interview. In stage 3, the feedback from the ten instructors was then analyzed and summarized into two main themes: monitoring students in VLs and grading students in VLs. Diverse needs were identified from the two themes. For monitoring, the following four requirements were identified:

- R1: Observing a single student in detail while having an overview of the entire class.
- R2: Showing addition class/student-based information.
- R3: Providing instructor-student communication channels.
- R4: Adjusting the monitoring scope.

For grading, the following two requirements were identified:

- R5: Applying different grading methods based on the task.
- R6: Allowing selective grading of lab tasks.

Based on the ideation step of the design thinking framework, corresponding features, which are described in the next section, are developed in stage 4. In stage 5, the features are included in the Grid-based preliminary prototype to create an improved prototype. In stage 6, we plan to invite domain experts in the experiment to review the improved prototype for appraising the feasibility of the monitoring tool and outline directions for improvements. We also plan to invite students to evaluation sessions to find out their needs and concerns, such as big-brother effects, for future development.

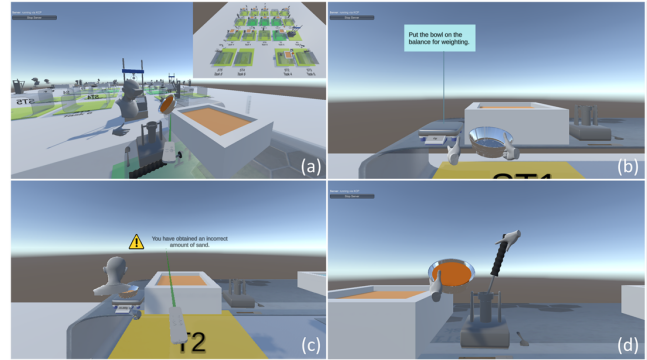


Figure 4: (a) Picture in picture view for simultaneous observation. (b) Text annotation in the student's environment. (c) Alert message shown after autograding. (d) First-person view for manual grading of a complex task.

3 SYSTEM PROTOTYPE

To address the requirements identified (R1-R6), the following features (F1-F6) are added to the Grid-based prototype (Figure 3) respectively to create the improved prototype:

- F1: A picture-in-picture view for monitoring. (Figure 4a).
- F2: Color-coded progress tiles and measurement task values (Figure 1).
- F3: Text annotation for student feedback (Figure 4b).
- F4: A filter to show a specific lab task.
- F5: Automatic grading of measurement tasks with an alert message (Figure 4c). Manual grading of complex tasks with first person view from the student's perspective (Figure 4d).
- F6: A grading policy to specify the tasks that require grading and the associated grading method.

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